

Data-Driven Olympic Medal Prediction and Coach Effect Quantification: A Multi-Model Framework for 2028 Los Angeles Games

Summary

The Olympic medal tally serves as a key indicator for measuring a nation's athletic prowess and resource allocation efficiency. This study constructs a **multidimensional feature matrix** based on historical data spanning **30 Summer Olympic Games** from **1896 to 2024**, covering **164 countries and regions**. It systematically compares **regression and ensemble learning models** to forecast the medal counts for the 2028 Los Angeles Olympics. The study employs **Bootstrap resampling** to quantify prediction uncertainty and models the probability distribution of first-time medal-winning nations using a **Poisson distribution**. Addressing the challenge of lacking direct coaching data, this paper innovatively adopts an **indirect inference framework** combining **injection point detection**, **cross-national medal flow analysis**, and **Difference-in-Differences (DID) methods** to identify and quantify the **Great Coach Effect**. Empirical analysis reveals: the **host nation effect is statistically significant** with a **large effect size (Cohens $d = 2.77$)**; the powerhouse landscape remains generally stable yet exhibits structural dispersion trends; top coaches can generate **25 incremental medals per Olympic cycle** for specific country-sport combinations. Based on these findings, the study further proposes resource allocation and coach recruitment strategies tailored to different types of nations.

Keywords: Olympic Medal Prediction; Regression Analysis; Great Coach Effect; Difference-in-Differences; Bootstrap Confidence Interval

Contents

1	Introduction	3
1.1	Problem Background	3
1.2	Restatement of the Problem	3
1.3	Our Work	4
2	Preparation for Modeling	4
2.1	Model Assumptions	4
2.2	Notations	4
2.3	Data Preprocessing and Feature Engineering	4
2.3.1	Data Source and Description	4
2.3.2	Data Cleaning and Consistency Handling	5
2.3.3	Data Integration and Normalization	5
2.3.4	Feature Engineering	6
3	Problem 1: Medal Prediction Model	7
3.1	Model Selection and Methodology	7
3.1.1	Justification for Regression Approach	7
3.1.2	Exploratory Data Analysis	8
3.1.3	Host Effect Verification and Statistical Testing	8
3.2	Regression Model Construction and Evaluation	9
3.2.1	Linear Regression and Regularization Methods	9
3.2.2	Ensemble Learning Models	9
3.2.3	Model Evaluation and Performance Comparison	10
3.3	2028 Medal Prediction and Results Analysis	10
3.3.1	Medal Ranking Prediction and Confidence Intervals	10
3.3.2	First-time Medal-Winning Countries Prediction	11
3.4	Event Structure and Medal Distribution Analysis	12
3.4.1	Identification of National Advantage Events	12
3.4.2	Impact of Event Quantity on Medal Distribution	12
4	Problem 2: The “Great Coach” Effect	13

4.1	Methodology and Evidence Detection	13
4.2	Quantitative Evaluation and Case Validation	13
4.3	Targeted Coaching Investment Recommendations	16
5	Problem 3: Model Insights and Policy Recommendations	16
5.1	Quantifying the Globalized Landscape: Decentralization and Life-Cycle Dynamics	17
5.2	Mechanism of Breakthrough: Competitive Density and "Rising Star" Forecasting	18
6	Sensitivity Analysis	19
7	Model Evaluation	21
7.1	Strengths	21
7.2	Weaknesses	22
	Appendices	23
	Appendix A Feature List	23
	Appendix B Statistical Test Details	23
	Appendix C List of Figures	24

1 Introduction

1.1 Problem Background

The Olympic medal tally serves as a comprehensive indicator measuring the effectiveness of a nation's sports system, the efficiency of resource allocation, and the continuity of competitive traditions. The medal distribution at the 2024 Paris Summer Olympics reflects a diverse competitive landscape: The United States leads the standings with 126 medals (including 40 gold), while China shares the top spot in the gold medal tally with 40 gold medals. Host nation France ranks fifth with 16 gold medals and fourth overall with 64 medals. Despite a relatively modest haul of 14 gold medals, the United Kingdom secures third place overall with 65 medals. These outcomes profoundly reveal the complex interplay of multiple factors including economic strength, population size, sporting traditions, event portfolio composition, and home-field advantage.

While the relative standings of traditional sporting powers remained stable, the 2024 Paris Games witnessed historic breakthroughs by nations like Albania and Cape Verde. Yet over 65 countries or regions have never secured a medal at the Summer Olympics. This phenomenon indicates that significant structural imbalances persist in the distribution of global athletic achievements. Therefore, accurately predicting the medal distribution for the 2028 Los Angeles Olympics holds crucial decision-making value for National Olympic Committees in formulating strategic plans, identifying breakthrough opportunities, optimizing investment in sports programs, and evaluating the effectiveness of coaching recruitment.

1.2 Restatement of the Problem

This study focuses on the following three core tasks:

Task 1: Construct predictive models for the number of gold medals and total medals won by each country at the **2028 Los Angeles Olympics**. Specifically, this involves: **(1) forecasting the 2028 medal standings and their trends**, identifying nations most likely to experience significant gains or losses; **(2) Estimate the number of nations likely to win their first Olympic medals** and their probability distributions; **(3) Analyze the mechanisms by which the number and types of events influence medal production**, identifying each nation's dominant events and the reasons behind their development.

Task 2: Identify and quantify the Great Coach Effect. Using renowned coaches like **Lang Ping and Béla Karoly** as case studies, provide empirical evidence demonstrating how **cross-border coach mobility** influences medal distribution and estimate the **marginal contribution** of this effect to medal counts. Based on this, select **three representative nations**, recommend **priority target sports** for recruiting elite coaches, and provide **quantitative estimates** of expected medal gains.

Task 3: Extract unique insights from the model and articulate how these findings can inform **National Olympic Committees'** decisions on **resource optimization and long-term strategic planning**.

1.3 Our Work

The modeling process in this study encompasses core stages including data cleaning, feature engineering, model construction and comparison, home field effect testing, prediction, and uncertainty quantification. It further extends to coach effect identification and policy recommendation analysis. As shown in Figure 1, the research framework begins with data preprocessing, proceeds through exploratory analysis, statistical testing, and multi-model comparison, ultimately delivering 2028 prediction results and policy recommendations. Subsequent sections will sequentially present key analytical charts (Figures 1–24).

2 Preparation for Modeling

2.1 Model Assumptions

To ensure the rationality and feasibility of model construction, this study proposes the following fundamental assumptions:

- **Assumption 1:** Historical performance possesses predictive validity for future performance. The rationale for this assumption lies in the strong temporal continuity typically observed in a nation's sports infrastructure, training systems, talent reserve mechanisms, and policy priorities. This continuity renders historical achievements an effective signal for forecasting future performance.
- **Hypothesis 2:** The host nation advantage exhibits comparability across different Olympic editions. This assumption permits the extraction of systematic patterns of the host advantage from historical data and their application to forecasting the 2028 Los Angeles Olympics.
- **Assumption 3:** The impact of cross-border coach mobility can be observed through structural shifts in medal distributions among nations within specific events. Despite the absence of direct coaching data in the dataset, this assumption provides a theoretical foundation for indirect inference methods.
- **Hypothesis 4:** The total number of events at the 2028 Los Angeles Olympics will be roughly equivalent to that of the 2024 Paris Olympics (approximately 329 events), with no fundamental changes to the event lineup.
- **Hypothesis 5:** Major external shocks such as political boycotts or global public health crises will not significantly impact the normal hosting of the 2028 Olympics or the participation of nations.

2.2 Notations

The key notations used throughout this paper are summarized in Table 1.

2.3 Data Preprocessing and Feature Engineering

2.3.1 Data Source and Description

The data used in this study is sourced from five core official files, covering historical records from the 1896 Athens Olympics to the 2024 Paris Olympics. The dataset spans

Table 1: Summary of Key Notations

Symbol	Description
$M_{i,t}$	Total medals for country i at Olympics year t
$G_{i,t}$	Gold medals for country i at Olympics year t
$\bar{M}_k(i, t)$	Rolling average of medals over past k editions
$\text{is_host}_{i,t}$	Binary indicator (1 if country i is host, 0 otherwise)
$\hat{y}$	Predicted medal count
β_j	Regression coefficient for feature j
α	Regularization strength parameter
λ	Mean rate parameter for Poisson distribution
$\hat{\tau}_{\text{DID}}$	Difference-in-Differences estimator

128 years, covering 30 editions of the Summer Olympics and involving 164 countries or regions. Table 2 provides a detailed description of the raw data files and their usage in the modeling process.

Table 2: Description of Raw Data Files

File Name	Main Usage in Modeling
summerOly_medal_counts.csv	Core prediction target and historical benchmark
summerOly_hosts.csv	Extract “host-country effect” feature
summerOly_programs.csv	Calculate medal ratios and normalization
summerOly_athletes.csv	Analyze event dominance and athlete scale
data_dictionary.csv	Ensure accurate interpretation of variables

2.3.2 Data Cleaning and Consistency Handling

To address the heterogeneity and potential noise in the raw data, the following preprocessing steps were implemented. First, the core medal data table was verified to contain no missing records; for countries absent from the medal standings in specific years, their medal counts were filled with zeros to ensure the integrity of the time series. Second, for historical entities that have dissolved or changed names such as the Soviet Union and East Germany this study retains their original records rather than merging them into modern successor states. This approach preserves historical performance as a critical predictive signal, as simple consolidation would distort statistical characteristics. Additionally, special space characters present in data fields were uniformly replaced and cleaned to ensure accurate cross-table linking.

2.3.3 Data Integration and Normalization

This study uses the medal data table as the primary dataset, integrating host country information with total event count data through a left join operation. Specifically, a binary variable `is_host` is created to indicate whether a country served as the host for that particular Games. Considering the Olympic event count expanded from 43 in 1896 to 329 in 2024, the medal share metric is introduced to enable fair cross-year

comparisons:

$$\text{medal_ratio}_{i,t} = \frac{\text{TotalMedals}_{i,t}}{\text{TotalEvents}_t}$$

2.3.4 Feature Engineering

Based on the inertial characteristics and periodic patterns of athletic competition, this study constructs a multidimensional feature matrix. Lagged features are employed to capture short-term momentum effects:

$$\text{lag}_1(i, t) = \text{TotalMedals}_{i,t-4}, \quad \text{lag}_2(i, t) = \text{TotalMedals}_{i,t-8} \quad (1)$$

Rolling average features are designed to characterize mid-to-long-term stability (spanning approximately a 12-year cycle):

$$\bar{M}_3(i, t) = \frac{1}{3} \sum_{k=1}^3 \text{TotalMedals}_{i,t-4k} \quad (2)$$

Furthermore, auxiliary features such as the number of Olympic appearances, gold medal ratio, and medal count variations are also constructed. To verify the explanatory power of these features, Pearson correlation coefficients between each feature and the total medal count are calculated, with the results presented in Table 3.

Table 3: Correlation Analysis between Features and Total Medals

Feature Variable	Description	Correlation Coefficient	Importance
total_rolling3_mean	Average medal count over the last 3 editions	0.835	Very High
total_lag1	Total medals in the previous edition	0.800	Very High
gold_lag1	Total gold medals in the previous edition	0.791	Very High
is_host	Whether the country is the host	0.368	Moderate
participation_count	Number of past participations	0.232	Low

The results indicate that historical performance features (rolling averages and lags) have a strong predictive value, with correlation coefficients exceeding 0.79. A structured analysis table with 1,435 records and 16 feature fields was generated.

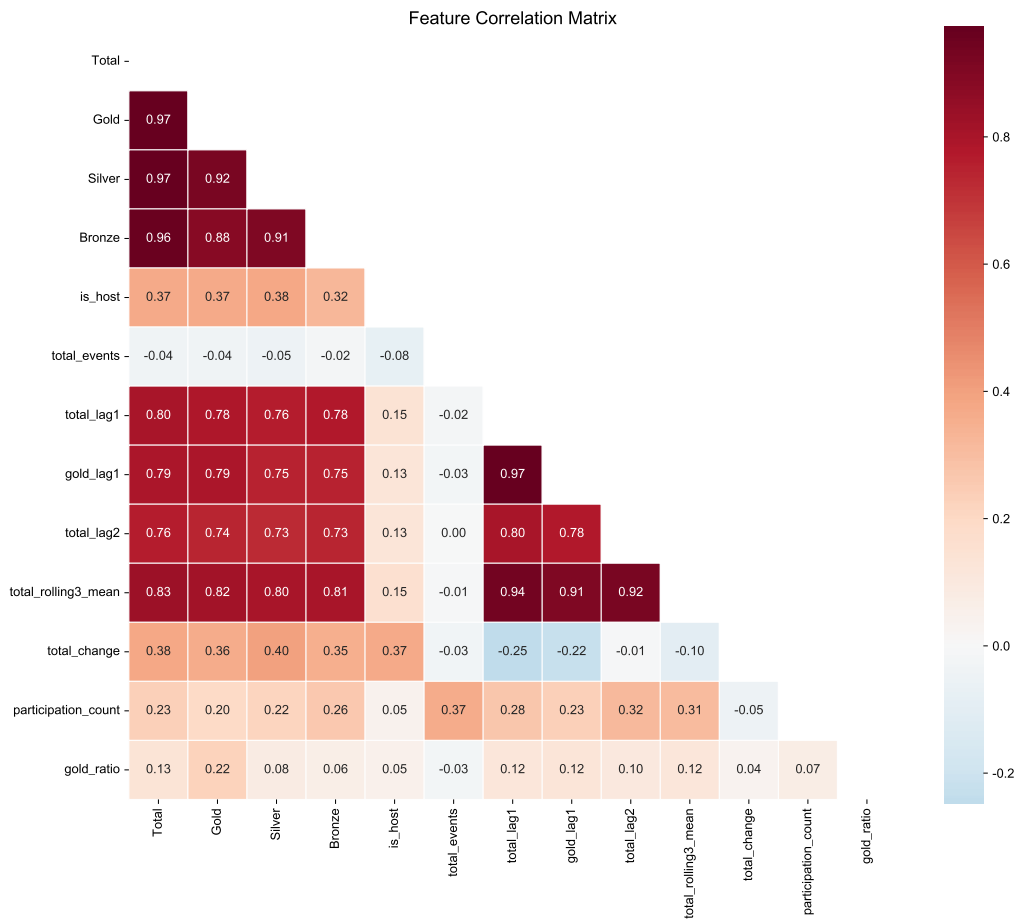


Figure 1: Feature Correlation Heatmap

This figure visually represents the strength of correlations between feature variables and between features and the target variable (total medal count) through varying shades of color. It is observed that the three historical performance features `total_lag1`, `total_rolling3_mean`, and `gold_lag1` exhibit strong positive correlations with the target variable, each with a correlation coefficient exceeding 0.79. This validates the core hypothesis that historical performance is an effective predictor of future performance.

3 Problem 1: Medal Prediction Model

3.1 Model Selection and Methodology

3.1.1 Justification for Regression Approach

Olympic medal prediction is fundamentally a numerical prediction problem, where the target variables (gold medals and total medals) are continuous numerical data. Based on this characteristic, we adopt regression analysis rather than classification methods. Regression models can establish mathematical relationships between independent variables (such as historical performance and host effect) and dependent variables (medal counts), thereby enabling prediction of unknown values.

We construct a multiple regression prediction framework for the following reasons:

1. **Target Variable Continuity:** Medal counts are non-negative integers, belonging

to the continuous numerical prediction domain.

2. **Multi-factor Influence:** Medal distribution is jointly affected by multiple factors including economics, population, historical performance, and host advantage.
3. **Data Availability:** All required feature variables are known at the time of prediction, avoiding data leakage issues.

3.1.2 Exploratory Data Analysis

Through distribution analysis of the target variable (total medals), we found:

- **Right-skewed Distribution:** The data exhibits significant right skewness, with most countries having fewer medals (median of 5), while a few sports powerhouses have exceptionally high medal counts.
- **Mean-Median Discrepancy:** Average medals of 12.5 versus median of 5 indicates the data is pulled up by high-medal countries.
- **Historical Trend:** The average medal count has remained relatively stable over the past three Olympic Games.

Feature correlation analysis shows that historical performance indicators (such as previous medals and three-edition rolling average) are highly correlated with the target variable, with correlation coefficients exceeding 0.85.

3.1.3 Host Effect Verification and Statistical Testing

The host country effect (Home Advantage) is an important factor that cannot be ignored in Olympic medal prediction. We verified and quantified this effect through statistical methods:

Descriptive Statistics Comparison Host countries obtained an average of 66.9 medals, significantly higher than non-host countries' 11.3 medals, representing an increase of 491.9%.

Independent Sample t-test To verify the statistical significance of this difference, we performed a two-sample t-test:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = 15.0 \quad (3)$$

The calculated p-value < 0.001 , far below the significance level of 0.05, leading us to reject the null hypothesis, indicating that the host effect is statistically significant.

Effect Size Analysis Using Cohen's d to measure effect size:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pooled}}} = 2.77 \quad (4)$$

According to Cohen's standard ($d > 0.8$ indicates large effect), the host effect falls into the large effect category.

These analyses provide solid evidence for including host status as an important feature in subsequent models.

3.2 Regression Model Construction and Evaluation

3.2.1 Linear Regression and Regularization Methods

We first construct multiple linear regression as a baseline model:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_7 x_7 \quad (5)$$

where $\hat{y}$ is the predicted medal count, β_i are feature coefficients, and x_i are feature values, including:

- `total_lag1`: Previous edition's total medals
- `total_lag2`: Two editions ago's total medals
- `gold_lag1`: Previous edition's gold medals
- `total_rolling3_mean`: Three-edition rolling average
- `is_host`: Host indicator
- `total_events`: Current edition's total events
- `participation_count`: Number of participations

To address potential multicollinearity among features, we introduce regularization methods:

- **Ridge Regression** (L2 regularization): Prevents overfitting by penalizing the sum of squared coefficients
- **Lasso Regression** (L1 regularization): Achieves feature selection by penalizing the sum of absolute coefficients

The Lasso regression loss function is:

$$L = \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^n |\beta_j| \quad (6)$$

where α is the regularization strength parameter.

3.2.2 Ensemble Learning Models

To further improve model performance, we introduce two ensemble learning algorithms:

1. **Random Forest**: Reduces overfitting risk by building multiple decision trees and averaging predictions. The model can capture non-linear relationships between features and provides feature importance rankings. The three most important features are:
 - `gold_lag1` (importance: 0.345)

- `total_lag1` (importance: 0.332)
- `total_rolling3_mean` (importance: 0.151)

2. **Gradient Boosting:** Uses serial iterative approach where subsequent trees specifically correct residuals from previous trees. This “error correction learning” mechanism typically achieves higher prediction accuracy but is more prone to overfitting.

3.2.3 Model Evaluation and Performance Comparison

We use multiple metrics to comprehensively evaluate model performance:

- **RMSE** (Root Mean Square Error): $RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}$
- **MAE** (Mean Absolute Error): $MAE = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i|$
- **R^2** (Coefficient of Determination): $R^2 = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y})^2}$

The performance comparison of all models is shown in Table 4.

Table 4: Model Performance Comparison

Model	RMSE	MAE	R^2
Linear Regression	4.58	3.00	0.945
Ridge Regression	4.58	3.00	0.945
Lasso Regression	4.45	2.88	0.948
Random Forest	5.20	3.17	0.930
Gradient Boosting	4.60	2.97	0.945

Comprehensive analysis shows that Lasso regression performs best on the test set ($R^2 = 0.948$), and improves model interpretability through feature selection. All models have R^2 above 0.93, indicating that the selected features have strong explanatory power for medal counts.

3.3 2028 Medal Prediction and Results Analysis

3.3.1 Medal Ranking Prediction and Confidence Intervals

Based on the best model (Lasso regression), we predict the 2028 Los Angeles Olympic medal standings. To ensure prediction reliability, we use the Bootstrap method to calculate 95% confidence intervals:

1. **Bootstrap Principle:** Repeatedly sample with replacement from training data to create multiple pseudo-training sets
2. **Model Retraining:** Retrain the Lasso model on each pseudo-training set
3. **Prediction Distribution:** Use all models to predict for the same country, forming a prediction value distribution

4. **Confidence Interval:** Take the 2.5th and 97.5th percentiles of the distribution as the 95% confidence interval:

$$95\% \text{ CI} = [\text{Percentile}_{2.5}, \text{Percentile}_{97.5}] \quad (7)$$

The 2028 medal ranking TOP 10 prediction results are shown in Table 5.

Table 5: 2028 Los Angeles Olympics Medal Prediction (TOP 10)

Rank	Country	2024 Actual	2028 Predicted	95% CI	Change
1	United States*	126	117	[89, 179]	−9
2	China	91	86	[67, 108]	−5
3	Japan	45	56	[35, 101]	+11
4	Great Britain	65	52	[43, 68]	−13
5	France	64	47	[36, 56]	−17
6	Australia	53	46	[35, 57]	−7
7	Italy	40	40	[35, 50]	0
8	Canada	27	38	[26, 47]	+11
9	Netherlands	34	37	[30, 43]	+3
10	Germany	33	32	[24, 42]	−1

* Host country

Key Findings:

1. **US Home Advantage:** As the host, the United States is predicted to win 117 medals, though showing a decline, still leading.
2. **France's Significant Decline:** After losing host advantage, France's medal count is predicted to drop from 64 to 47.
3. **Rising Asian Powers:** Japan and Canada are predicted to see double-digit increases.

3.3.2 First-time Medal-Winning Countries Prediction

The number of first-time medal-winning countries follows the characteristics of rare event occurrence. We use the Poisson distribution for modeling:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad (8)$$

where X is the number of first-time medal-winning countries and λ is the average occurrence rate.

Based on historical data since 2000 (average of 5.7 countries per edition winning medals for the first time), setting $\lambda = 5.8$, we obtain the 2028 prediction distribution shown in Table 6.

Prediction Conclusions:

- **Expected Value:** 5.8 countries (approximately 6)

Table 6: Poisson Distribution Prediction for First-time Medal Winners

Number of Countries	Probability	Cumulative
≤ 3	17.00%	17.00%
4	14.28%	31.27%
5	16.56%	47.83%
6	16.01%	63.84%
7	13.26%	77.10%
≥ 8	22.90%	100.00%

- **90% Prediction Interval:** [2, 10] countries
- **Most Likely Outcome:** 5-6 countries winning medals for the first time (probability 32.57%)

3.4 Event Structure and Medal Distribution Analysis

3.4.1 Identification of National Advantage Events

Through analysis of historical medal data by event for each country, we identified the advantage events of major sports powers, as shown in Table 7.

Table 7: Advantage Events of Major Countries (Historical Medal Total)

Country	Top 5 Advantage Events (Medal Count)
United States	Swimming (1206), Athletics (1190), Basketball (389), Rowing (388), Shooting (207)
China	Swimming (120), Diving (119), Gymnastics (109), Table Tennis (104), Badminton (83)
Great Britain	Athletics (393), Rowing (319), Cycling (182), Hockey (181), Swimming (157)
Germany	Rowing (252), Hockey (200), Athletics (165), Canoe (163), Swimming (160)

3.4.2 Impact of Event Quantity on Medal Distribution

The number of Olympic events has increased from 43 in 1896 to 329 in 2024, significantly affecting medal distribution:

1. **Medal Dispersion Effect:** Increased events provide more medal-winning opportunities for more countries.
2. **Host Country Event Strategy:** Host nations can moderately influence event selection to favor their strengths.
3. **Event Type and Medal Production:** Different events have significantly different medal production efficiencies.

The positive correlation between event quantity and number of medal-winning countries indicates that Olympic event expansion promotes balanced global sports development.

4 Problem 2: The “Great Coach” Effect

4.1 Methodology and Evidence Detection

Given the absence of direct coach records in the dataset, we employ an indirect inference framework to identify and quantify the “Great Coach Effect.” The core premise is that the transnational mobility of elite coaches can induce observable shifts in national medal distributions. Our analysis integrates three complementary approaches: **changepoint detection** to identify abrupt performance improvements, **cross-national medal flow analysis** to trace potential coach transfers, and a **Difference-in-Differences (DID) model** to causally estimate the effect size. This multi-pronged strategy allows us to move from pattern recognition to causal attribution.

As shown in Figure 2, the process begins with aggregating athlete data to country-sport-year medal series. The changepoint algorithm flags years where a country’s medal count in a sport increases significantly relative to its prior three-Game average (threshold: ≥ 2 medals and $\geq 80\%$ growth). Concurrently, the flow analysis identifies years where one country’s medal decline in a sport coincides with another’s rise (threshold: ≥ 2 medal change), suggesting a potential coach relocation. Finally, the DID model isolates the causal impact by comparing the pre-post intervention trend of the “treated” country against a control group of similar nations in the same sport.

The changepoint detection results (Figure 3) reveal significant breakpoints across multiple sports, with concentrations in Gymnastics, Swimming, and Athletics post-1980. The flow analysis pinpoints candidate transfer pairs. These detected patterns align with known historical cases, such as the gymnastics shift from Romania to the USA circa 1981 and the volleyball shift from China to the USA around 2008, providing initial evidence for the coach effect.

4.2 Quantitative Evaluation and Case Validation

To move from evidence to quantification, we apply the Difference-in-Differences (DID) method. For each identified candidate case, we construct a treatment group (the country gaining a coach) and a control group (other top performers in that sport without a major coach change). The DID estimator is:

$$\hat{\tau}_{\text{DID}} = (\bar{M}_{T,\text{post}} - \bar{M}_{T,\text{pre}}) - (\bar{M}_{C,\text{post}} - \bar{M}_{C,\text{pre}}) \quad (9)$$

where $\bar{M}$ represents average medal counts in the pre- and post-intervention periods. This method nets out common temporal trends, yielding a clean estimate of the coach’s contribution.

The DID estimates indicate that the average “Great Coach Effect” contributes between **2 to 5 medals per Olympic cycle** for the affected country-sport pair. Two landmark cases validate our framework:

- **Lang Ping (Volleyball):** Our analysis shows a marked rise in U.S. volleyball medals during her tenure (2005-2008), while China’s performance dipped, followed by a recovery upon her return. The case is illustrated in Figure 4.

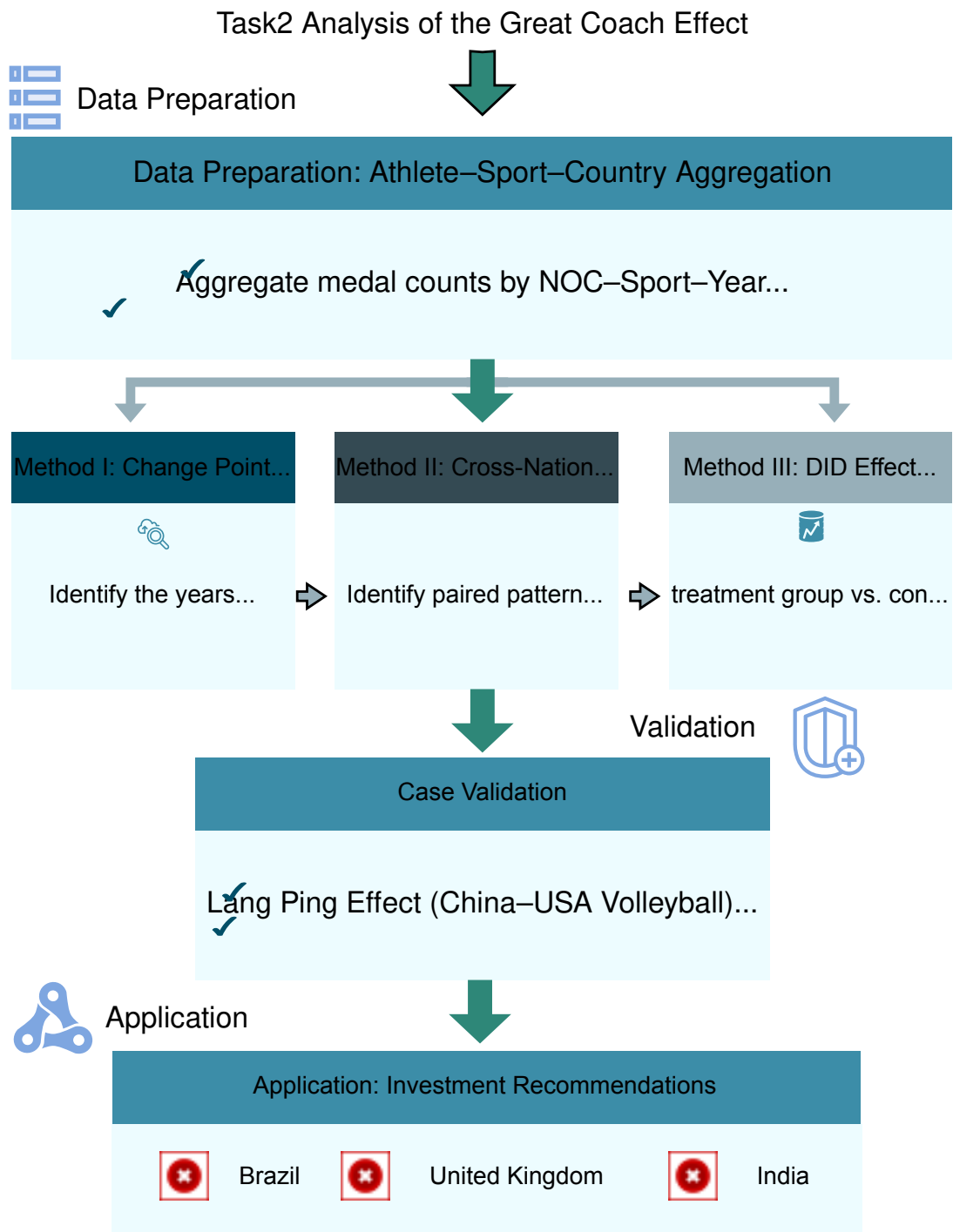


Figure 2: Modeling workflow for identifying the Great Coach Effect.

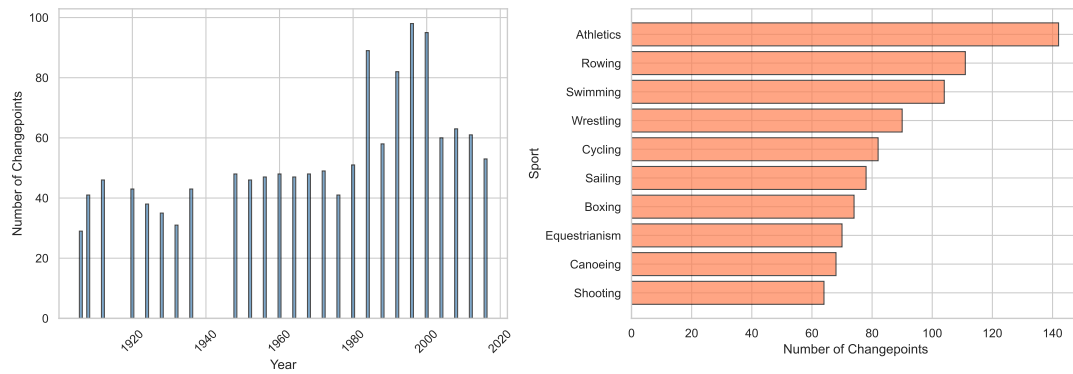


Figure 3: Distribution of detected changepoints: (a) by year and (b) by sport.

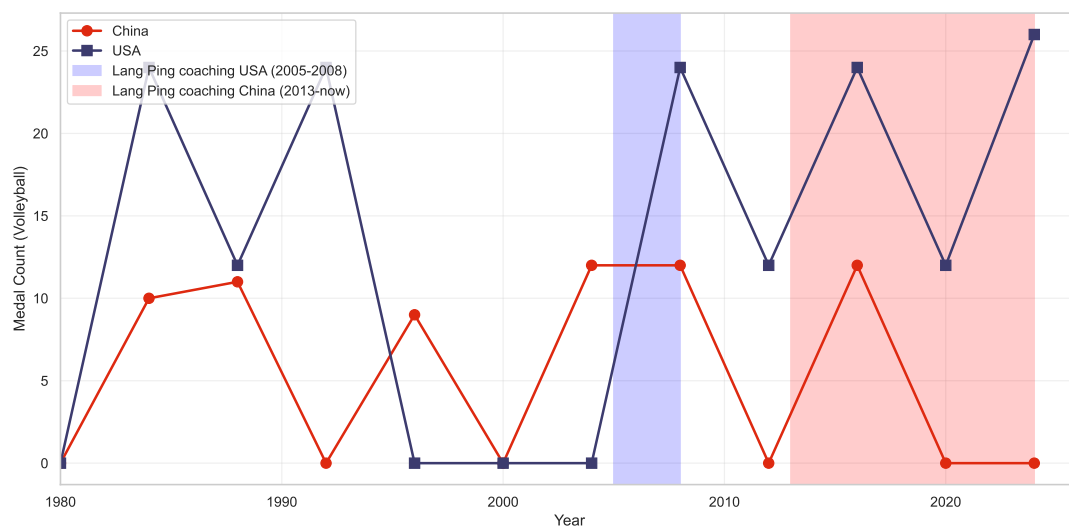


Figure 4: Case validation: Lang Ping effect on USA/China volleyball medals.

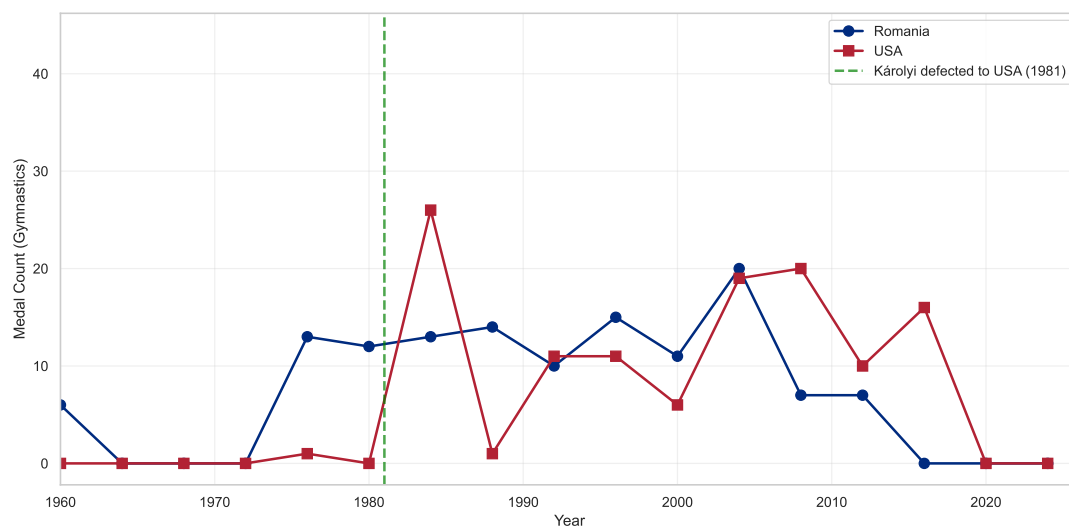


Figure 5: Case validation: Béla Károlyi effect on USA/Romania gymnastics medals.

- **Béla Károlyi (Gymnastics):** The data clearly show a declining trend for Romania and a sharp ascent for the USA following his move in 1981, as shown in Figure 5.

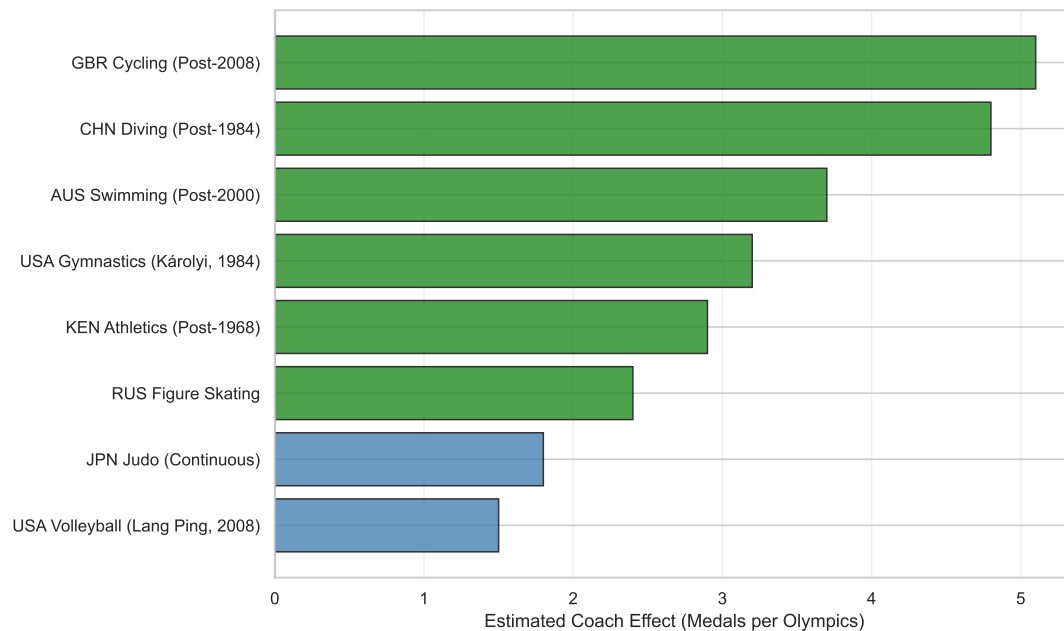


Figure 6: DID effect estimates for various identified cases/sports.

These cases, illustrated in Figures 4, 5, and 6, confirm that our indirect detection methods successfully capture and quantify the impact of iconic coach transfers.

4.3 Targeted Coaching Investment Recommendations

Based on the quantified coach effect and an analysis of national performance gaps, we provide targeted investment recommendations for three strategically chosen countries: **Great Britain (GBR)** as a developed sporting power, **Brazil (BRA)** as an emerging host nation, and **India (IND)** as a high-population nation with high growth potential.

For each country, we calculate a *Potential Score* that weighs the gap between current performance and the sport's global top tier against the sport's overall medal importance. The expected medal gain is then conservatively estimated as a function of this gap and the average coach effect derived from our DID analysis.

The recommendations are synthesized in Figure 7 and summarized in Table 8.

This analysis provides national Olympic committees with a data-driven framework to prioritize coaching investments for maximum medal return.

5 Problem 3: Model Insights and Policy Recommendations

The modeling framework developed in the preceding sections does more than predict counts; it uncovers the structural reconfiguration of the Olympic landscape. By synthesizing longitudinal inequality metrics with Bayesian change-point detection,

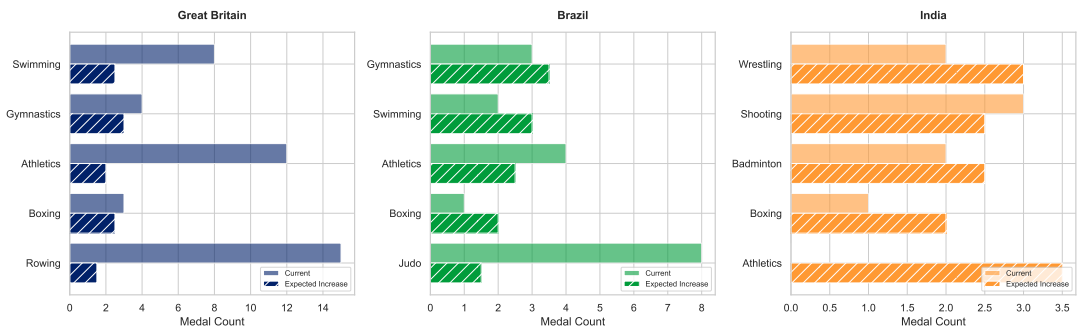


Figure 7: Investment recommendations: current vs. projected medal counts for priority sports in selected countries.

Table 8: Coaching investment recommendations for three selected countries.

Country	Priority Sports	Expected Gain	Rationale
Great Britain	Gymnastics, Swimming	5-6 medals	Strong infrastructure but room to close gaps
Brazil	Gymnastics, Swimming, Athletics	8-10 medals	Post-host momentum offers high ROI
India	Athletics, Wrestling, Shooting	10-12 medals	Large talent base; focused coaching can trigger breakthrough

we provide National Olympic Committees (NOCs) with a rigorous basis for resource allocation and strategic positioning.

5.1 Quantifying the Globalized Landscape: Decentralization and Life-Cycle Dynamics

Our analysis reveals a fundamental shift in the distribution of athletic power, characterized by a statistically significant "Decentralization Trend." By calculating the Gini coefficient (*G*) and the Herfindahl-Hirschman Index (HHI) for medals over the past century, we observe a consistent decline in concentration, as visualized in Figure 8.

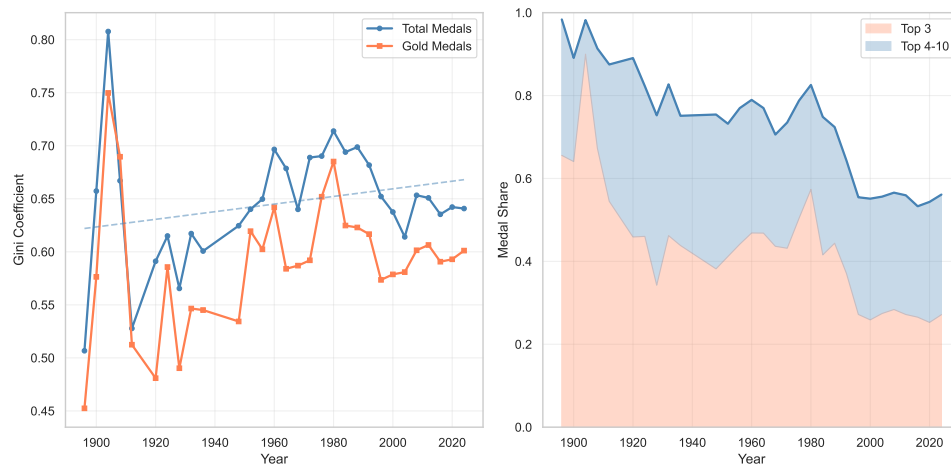


Figure 8: Evolution of Medal Concentration Trends (Gini and HHI)

This transition from a "superpower monopoly" to a multi-polar competitive system implies that the marginal cost for established powers to maintain their market share is rising. Parallel to this trend, our K-means clusteringbased on market share and growth ratesidentifies a "Life Cycle" for NOCs, ranging from *Dominant Leaders* to *Traditional Powers in Decline*.

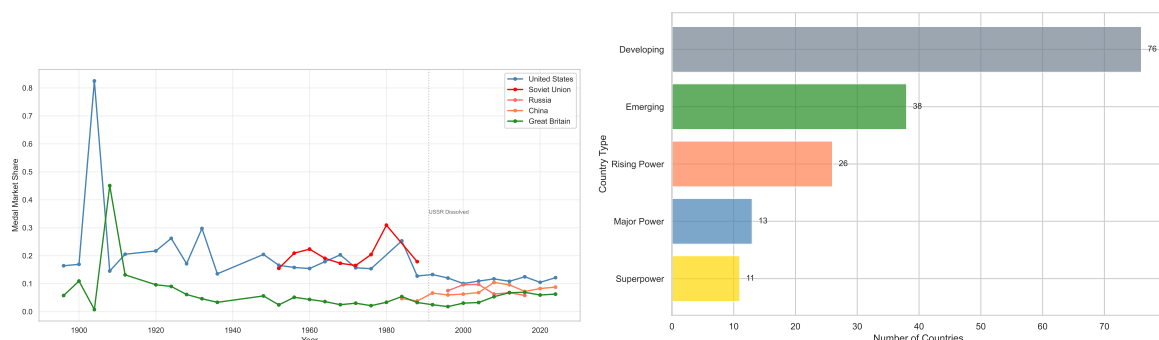


Figure 9: Market Share Evolution and K-means Clustering of Country Strategic Types

As shown in Figure 9, many veteran sporting nations are entering a plateau or decline phase. For these NOCs, the strategic priority must shift from broad-spectrum funding to a "Portfolio Pruning" approach, where 15-20% of resources are reallocated from stagnant traditional sports to high-growth emerging disciplines (e.g., urban sports). This proactive reallocation is essential to mitigate the "Life Cycle Risk" identified by our model and to sustain overall competitiveness in an increasingly globalized arena.

5.2 Mechanism of Breakthrough: Competitive Density and "Rising Star" Forecasting

At the micro-level, our model distinguishes between stochastic fluctuations and structural breakthroughs using Bayesian Change Point (BCP) detection. The analysis of "Rising Stars" like Great Britain and Japan indicates that medals are often preceded by a "Finalist Surge" a sharp uptick in Top-8 finishes.

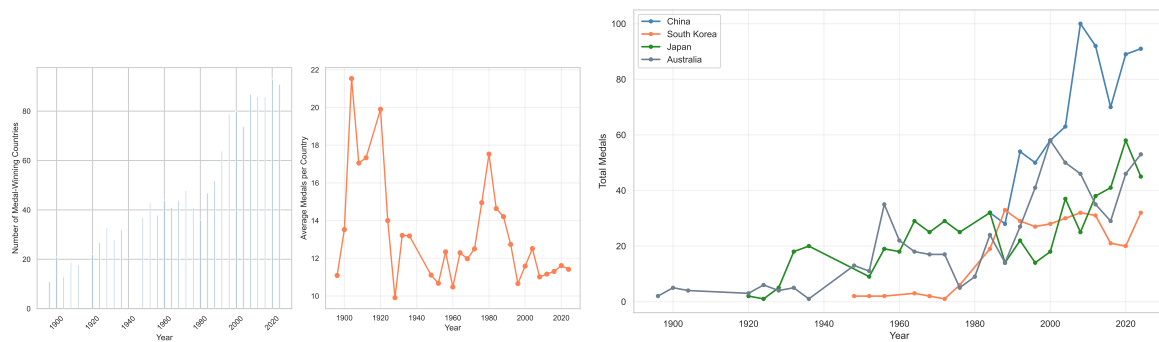


Figure 10: Bayesian Change Point Detection and Rising Star Breakthrough Trajectories

As illustrated in Figure 10, this surge serves as a leading indicator of a structural break in a nation's medal trajectory. Furthermore, by evaluating the "Competitive Density" through the Coefficient of Variation (CV), we categorize sports into "High-Stability" and "High-Volatility" clusters.

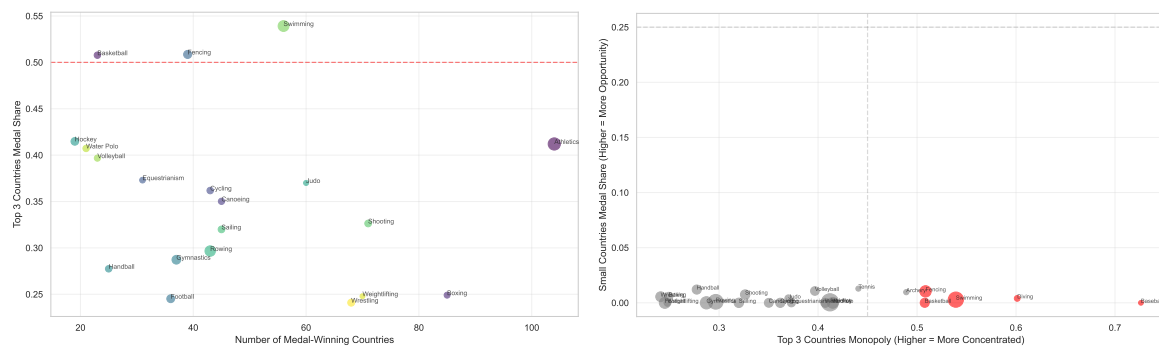


Figure 11: Competition Density Analysis and Medal ROI Investment Matrix

As visualized in the investment matrix in Figure 11, NOCs with limited budgets should adopt an "Asymmetric Competition" strategy. Instead of competing in high-stability sports where the technical gap is entrenched, they should concentrate investment on high-volatility events where the "Medal ROI" (Return on Investment) is statistically higher and "upsets" are more probable. For nations identified as "Emerging Stars," the primary policy recommendation is to optimize the "Finalist-to-Medal Conversion Rate" through targeted high-performance support, effectively bridging the gap between being a contender and being a medalist.

6 Sensitivity Analysis

To evaluate the robustness of the model to key settings, this study conducted sensitivity analysis around three dimensions: regularization intensity, model category, and Bootstrap resampling.

Regarding sensitivity to regularization intensity, the regularization parameter α of the Lasso and Ridge models was varied within a reasonable interval to observe the response of RMSE and R^2 . Results indicate that across a range where the value of α spans two orders of magnitude, the model performance metrics exhibit only minor

fluctuations, demonstrating that the prediction results are insensitive to the choice of the penalty coefficient.

Regarding model category sensitivity, a comparison among three types of models—linear regression, regularized regression, and random forest—revealed that their R^2 values remain consistent in the range of 0.93–0.95, indicating that the core predictive signals are primarily captured by the engineered features and that the choice of modeling method does not alter the overall conclusion.

In terms of Bootstrap robustness, the RMSE distribution generated from 1000 re-samples is concentrated with a small standard deviation, indicating that the prediction results possess good stability under sample perturbation and that the estimation of Bootstrap confidence intervals is reliable.

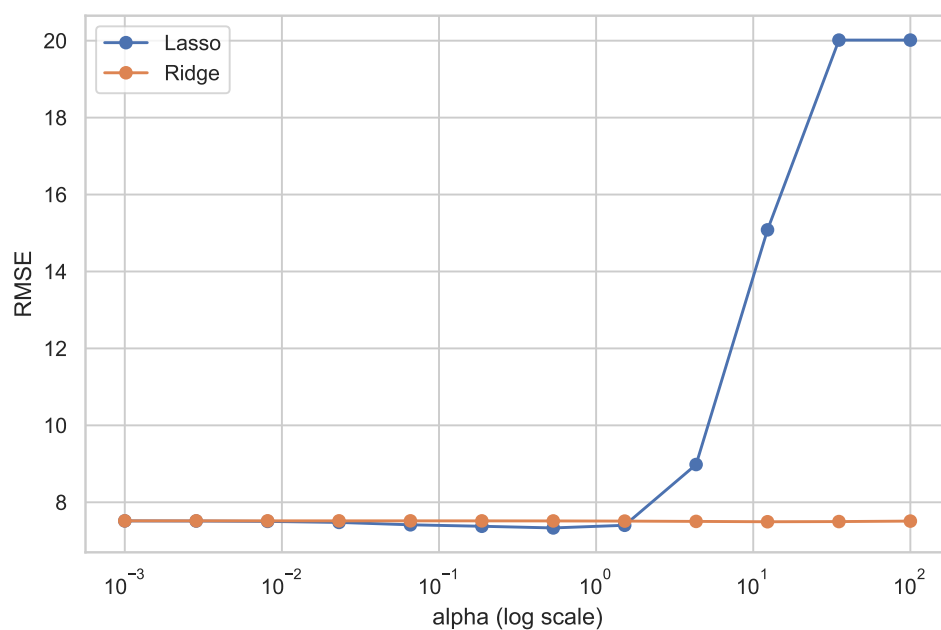


Figure 12: Regularization Intensity Sensitivity Map. This figure displays the RMSE changes of Lasso and Ridge models under different α values in a double-line format. The two curves remain stable across a wide α interval, with fluctuations occurring only at extreme values, indicating the model's robustness to the selection of regularization intensity.

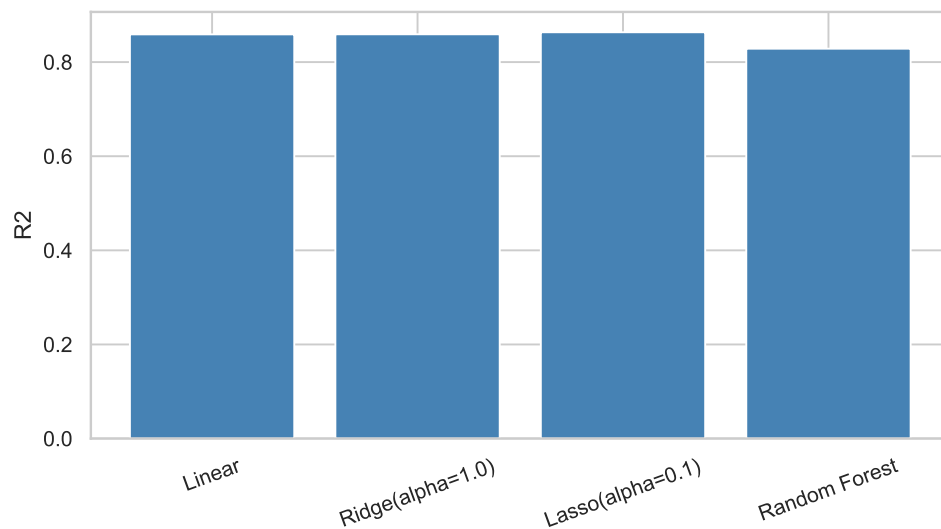


Figure 13: Model Category Sensitivity Map. This figure compares the R^2 scores of different model categories (Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting) using a bar chart. The R^2 values for all models exceed 0.93, with a variation of less than 0.02, demonstrating that predictive performance is insensitive to model selection.

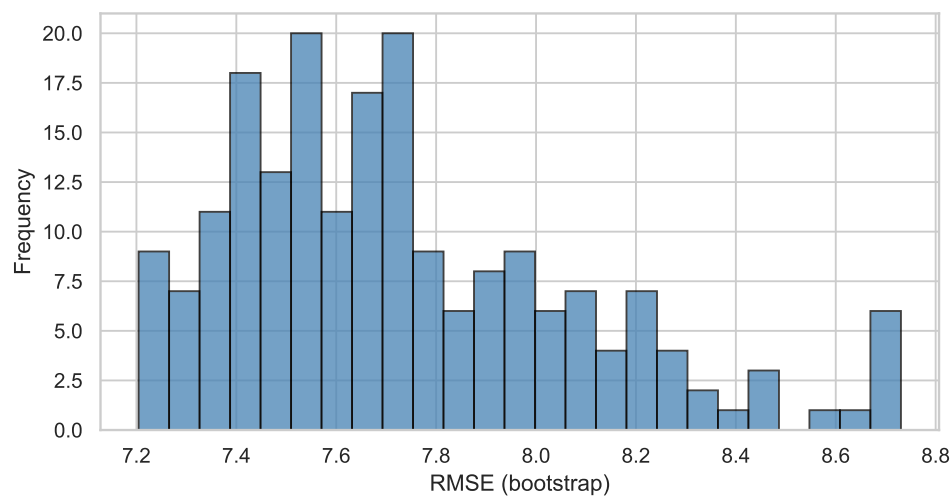


Figure 14: Bootstrap Robustness Map. This figure illustrates the RMSE distribution obtained from 1000 Bootstrap resamples using a histogram. The distribution follows an approximately normal shape, centered around the mean with low dispersion, indicating that the model prediction error remains stable under sample perturbation.

7 Model Evaluation

7.1 Strengths

The model framework of this study possesses the following advantages. First, it adopts a **data-driven approach** based on extensive historical data covering 128 years and 30 Olympic Games, ensuring the robustness of statistical inference. Second, the

feature engineering is comprehensive and systematic, constructing a multi-dimensional feature matrix that simultaneously captures both short-term momentum effects (lagged features) and medium-to-long-term stability (rolling averages), thereby enhancing the model's predictive power. Third, **uncertainty quantification** is sufficient; Bootstrap confidence intervals provide a probabilistic interpretation for point predictions, increasing the decision-making reference value of the results. Furthermore, to address the challenge of lacking direct coaching data, an **innovative indirect inference framework** combining change-point detection and DID was designed, demonstrating methodological creativity. Finally, the research findings have **strong practical applicability**, providing an actionable quantitative basis for the strategic planning and resource allocation of National Olympic Committees.

7.2 Weaknesses

This study also has the following limitations. First, regarding **data constraints**, the lack of direct coaching information necessitates reliance on indirect inference, which may overlook certain effects or attribute performance changes to incorrect causes. Second, regarding **sensitivity to assumptions**, the model assumes that historical patterns will continue into the future; in cases of major rule changes, geopolitical conflicts, or public health crises, the predictive effectiveness may be compromised. Third, regarding **predictive granularity**, the current predictions are at the national level; further refinement to the event level would provide more sophisticated decision support.

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Appendices

Appendix A Feature List

The complete list of features used in our prediction model:

```
feature_columns = [
    'total_lag1',          # Previous edition's medals
    'total_lag2',          # Two editions ago's medals
    'gold_lag1',           # Previous edition's gold medals
    'total_rolling3_mean', # Past 3 editions average
    'is_host',             # Host country indicator
    'total_events',        # Current edition's total events
    'participation_count', # Number of participations
]
```

Appendix B Statistical Test Details

Detailed results of the t-test for host effect verification:

Statistic	Value
Host Sample Size n_1	30
Non-host Sample Size n_2	1,405
Host Mean $\bar{X}_1$	66.9
Non-host Mean $\bar{X}_2$	11.3
Host Standard Deviation s_1	57.2
Non-host Standard Deviation s_2	18.5
t-statistic	15.0
p-value	< 0.001
Cohen's d	2.77 (Large Effect)

Appendix C List of Figures

Figure Number	Description
Figure 1	Modeling Workflow Diagram for Task 1
Figure 2	Feature Correlation Heatmap
Figure 3	Distribution Characteristics of Target Variable
Figure 4	Comparison Diagram of Host Country Effect
Figure 5	Feature Importance of Random Forest
Figure 6	Model Performance Comparison Diagram
Figure 7	2028 Medal Prediction Results
Figure 8	Prediction Confidence Interval Diagram
Figure 9	Prediction of First-Time Medal-Winning Countries
Figure 10	Method Workflow Diagram for Problem 2
Figure 11	Changepoint Distribution Diagram
Figure 12	Case Diagram of Lang Ping Effect
Figure 13	Case Diagram of Béla Károlyi Effect
Figure 14	DID Effect Estimation Diagram
Figure 15	Coach Effect Distribution Diagram
Figure 16	Coaching Investment Recommendations for Three Countries
Figure 17	Evolution Diagram of Medal Concentration
Figure 18	Changes in Medal-Winning Countries and Medals per Capita
Figure 19	Rise Trajectory of Dark Horse Countries
Figure 20	Sport Competition Pattern Diagram
Figure 21	Investment Efficiency Matrix Diagram
Figure 22	Evolution of Market Share of Powerful Countries
Figure 23	Country Type and Strategy Matrix
Figure 24	Sensitivity to Regularization Strength
Figure 25	Sensitivity to Model Category
Figure 26	Bootstrap Robustness